

ORIGINAL ARTICLE

Unlocking renewable energy potential: A case study of solar and wind site selection in the Kasserine region, central-western Tunisia

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Abstract

Developing wind and solar photovoltaics on a large scale requires substantial financial investments, making it crucial to identify the most suitable locations beforehand. To address this issue, a spatial analysis is carried out to determine the most potential sites for hosting large-scale solar photovoltaic and wind systems in the region of Kasserine, central-western Tunisia. To this end, an integrated model based on Step-wise Assessment Ratio Analysis (SWARA), Decision-Making Trial and Evaluation Laboratory (DEMATEL), and Geographic Information System is proposed. An extensive literature survey is conducted to establish suitability criteria and constraints. The SWARA-DEMATEL is used to assign weights and capture the interdependencies among the considered criteria, while the raster calculator tool from the ArcGis 10.8 software is utilized to extract the final suitability maps. The obtained results indicate that the region of Kasserine exhibits great solar and wind potential, with areas of 635 and 467 km² extremely fit for installing solar and wind systems, respectively. Furthermore, 349 km² are identified as potential locations for hosting solar-wind hybrid systems. Considering these outcomes, policymakers can take the initiative to accelerate the deployment of these facilities, which would assist the country in achieving its plans by 2030.

KEYWORDS

decision-making trial and evaluation laboratory, Geographic Information System (GIS), multicriteria decision-making, solar and wind energy, step-wise assessment ratio analysis, Tunisia

1 | INTRODUCTION

Renewable energy has become a critical topic of discussion in recent years as societies worldwide increasingly recognize the importance of transitioning to sustainable energy sources. Furthermore, these

renewable energy sources (RES) have distinct advantages over conventional fuels, which are subject to price volatility and geopolitical tensions.¹ RES are often domestically available and can contribute to energy independence. Additionally, developing and utilizing RES can stimulate economic growth and

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create new job opportunities, which helps spur technological advancements and foster a sustainable and resilient economy.^{2,3}

Hence, to secure energy supplies, reduce energy dependency and costs, and consider the serious environmental impacts, many governments have embraced their policies for utilizing these RES.^{4–7} Driven by impressive technological enhancements and remarkable cost declines, the renewable energy sector, solar photovoltaic (PV) and onshore wind in particular, has seen remarkable growth in recent years, attracting investments and driving innovation. Which is why both sources have been regarded as the most viable alternatives for a green transition.^{8–11} With newly added capacities of 191 and 75 GWh/year for solar PV and wind, respectively, both technologies have taken the lead in terms of fast-growing technologies.¹²

On a national scale, as a net importer, Tunisia is faced with a multitude of challenges concerning its energy system. The soaring demand and dwindling local reserves have led to a chronic deficit. The power system is almost entirely based on natural gas (97%), which indicates that the energy mix does not exist, as the share of renewables is slightly less than 3%, including solar, wind, and hydro.^{13–15} The current installed capacity stands at 5944 MW, with 92% run by the national state company (STEG) and 8% by the private Carthage Power Company (CPC).¹³ Moreover, the STEG monopoly, reliance on imported natural gas, and heavy subsidies all further exacerbate the sector as a whole.¹⁶ Nevertheless, given its geographical location and meteorological characteristics, Tunisia is well-endowed with significant solar and wind potential. From a statistical perspective, vast parts of the country enjoy wind speeds of 6–8 m/s, while global horizontal irradiance varies between 4.93 and 7.12 kWh/m²/day.^{17,18} Several reports have elaborated on the technical potential of solar PV, solar CSP, onshore, and offshore wind as 400, 65, 10, and 250 GW, respectively.^{14,15} In this regard, the Tunisian government has adopted an ambitious plan, the Tunisian Solar Plan (TSP), vowing to install 4.7 GW of renewable energy by 2030 to secure its energy supplies, diversify its power system, and reduce its imports.¹⁹

To effectively utilize solar and wind energy sources, it is crucial to identify suitable sites for the installation of such systems. Site selection plays a vital role in unlocking the potential of renewable energy projects. However, identifying a well-suited geographical location for solar and wind installations requires a comprehensive understanding of various factors, including topography, climate conditions, infrastructure availability, and environmental impacts.^{20–22} Thus, it is critical to decide on the

most appropriate site before developing solar and wind systems.

While existing studies have already explored site selection for renewable energy installations using various methodologies, there is always room for further research for several reasons, as each region has its own unique geographical, climatic, and socioeconomic characteristics. It has been noted from the extensive literature that several studies have been dedicated to exploring opportunities for utilizing RES in Tunisia. Such studies focused on determining solar (PV and CSP) and wind (onshore and offshore) potential at a national and regional level.^{17,21–24} Nevertheless, the region of Kasserine, with its specific conditions, has been overlooked in previous studies regarding the implementation of solar PV and wind turbine-based power generation systems. Being labeled as the most marginalized region of the country with the highest national poverty and unemployment rates, identifying optimal sites for installing solar PV and wind projects would not only ensure a stable power supply but also have a profound impact on job creation. This would spur economic growth through increased investments in renewable energy infrastructure. Therefore, the main endeavor of this study was to provide tailored insights into this area, contributing to more precise and effective planning for local renewable energy projects. The novelty of this study lies in its integrated approach combining Step-wise Assessment Ratio Analysis (SWARA), Decision-Making Trial and Evaluation Laboratory (DEMATEL), and Geographic Information System (GIS). While each method has been used individually or in different combinations in other studies, this particular integration may offer a new perspective on assessing interdependencies among criteria and provide a more robust decision-making framework for site selection. To further address the current research gap, the following key questions are sought to be answered:

- Q1: What are the socioeconomic outcomes of installing solar and wind systems in the Kasserine region, particularly related to job creation, energy costs?
- Q2: How do the geopolitical and legislative environments of Tunisia influence the practical deployment of these systems in the identified sites?
- Q3: Can the adopted approach for the Kasserine region serve as a model for other similar regions in Tunisia and other countries?

This current study focuses on the Kasserine region, central-western Tunisia, using a GIS-based SWARA-DEMATEL integrated approach with a view to determining the most feasible locations for deploying solar PV and

onshore wind facilities on a large scale. It is anticipated that the adopted approach would provide valuable insights and assist in making informed site selection decisions, which would contribute to the overall goal of unlocking the renewable energy potential in the region.

2 | LITERATURE REVIEW

Site selection plays a crucial role in the success of solar and wind energy projects. However, it is not without its challenges. Understanding these challenges is essential to unlocking the renewable energy potential in any given region. It is insufficient to develop a solar or wind facility based merely on the abundance of solar and wind sources, as there are many other factors that have a direct impact on the output power, such as the topography of the location, infrastructure, and costs.^{25–27} In addition, solar and wind potential vary significantly from one region to another according to weather patterns and seasonal variations.²⁰ Therefore, collecting and analyzing data on a potential site is crucial to determine its energy generation capacity.^{20,26,28} However, this is not a straightforward task in light of the various and often overlapping criteria illustrated in Table 1. In this context, the integration of GIS along with MCDM approaches has emerged as a valuable tool in handling decision-making complexities, including solar and wind site selection.^{21,43,44}

From the literature, it is clear that many researchers have successfully used a variety of MCDM techniques to address the issue of choosing the most practical geographic locations for installing solar and wind facilities. Among the most frequently used methods are Analytic Hierarchy Process (AHP),^{20,37,45} Fuzzy AHP (FAHP),^{46,47} Technique for Order Preference by Similarity to Ideal Solution (TOPSIS),⁴⁸ Data Envelopment Analysis (DEA),⁴⁹ Multicriteria Analysis System applied as a Baseline Method to Environmental Impact Assessment (SAMAMBAIA),⁵ and many other methods. Yet, recently, SWARA and DEMATEL techniques have been increasingly used due to their advantageous features over the other models. As an expert-oriented model, SWARA was developed by Zolfani and Saparauskas⁵⁰ to assess the sustainability of an energy system. In Iran, the same approach was used by Vafaeipour et al.⁵¹ to investigate the feasibility of solar projects in 25 cities. In a very recent study, Yücenur and Ipekçi⁵² integrated SWARA with WASPAS to identify a suitable location for a marine power plant.

To capture and analyze the interdependencies among the multiple factors related to the decision-making problem, the DEMATEL approach was employed in

TABLE 1 Frequently used criteria for developing solar PV and wind facilities in the literature.

Criteria	Reference
Global Horizontal Irradiance (GHI)	[25, 26, 29–31]
Average Wind Speed	[25, 26, 29, 30, 32]
Wind power density	[26, 33–35]
Percentage of windy days	[33]
Ambient temperature	[31, 36–38]
Slope	[26, 31, 32, 34, 36]
Aspect	[20, 37, 39]
Distance to grid lines	[20, 21, 26, 31, 32]
Distance to major roads	[20, 21, 26, 31, 32]
Distance to residential areas	[20, 21, 26, 31, 32]
Elevation	[21, 25, 29, 30, 32]
Land use	[21, 25, 29–32]
Population density	[39, 40]
protected Bird areas	[25, 32, 41]
Distance to airports	[25, 29, 32, 41]
Dust storm	[42]
Distance to water resources	[25, 37, 38]

several studies. For example, Azizi et al.³³ developed a GIS-based DEMATEL approach to determine the criteria interrelationships for wind site selection in Iran. Büyüközkan and Güleriyüz⁵³ proposed an integrated DEMATEL-ANP approach to select the most appropriate RES in Turkey. In recent works, the classical DEMATEL has been used to explore the various barriers pertinent to implementing RESs in emerging economies.^{54–56} Taking advantage of both techniques, Badi et al.²¹ combined SWARA and DEMATEL along with GIS to determine the most potential locations for deploying solar power plants in Misurata district, western Libya. The authors stressed that the adopted model was very effective and outperformed other similar weighting techniques like the Best Worst Method (BWM) and AHP.

Integrating GIS with the SWARA-DEMATEL has proven to be an innovative model that provides several advantages that contribute to more efficient and adequate decision-making in renewable energy projects. By utilizing the GIS-based SWARA-DEMATEL approach, developers can identify sites with optimal solar and wind resources, minimizing the need for costly transmission infrastructure and maximizing the energy generation potential. This not only reduces the initial investment but also leads to long-term operational savings, making renewable

energy projects economically viable. Therefore, this research work makes significant contributions to the existing literature in several ways. First, it presents an original approach to developing criteria layers for large-scale wind and solar PV power plants using GIS tools combined with the SWARA-DEMATEL model. Second, it examines the opportunities for deploying solar and wind, as well as solar-wind hybrid systems, in the governorate of Kasserine, which is known for being the most marginalized region in the country. Therefore, this study is unique as it represents the first comprehensive evaluation of both resources for potential deployment in this designated region, which can serve as a valuable reference for future investigations regarding spatial analyses of such technologies.

3 | METHODOLOGY

Site selection for solar and wind energy projects is a complex process that involves addressing challenges related to land availability, resource assessment, environmental and social impacts, and infrastructure requirements. Therefore, to unlock the renewable energy potential in any given region, it is critical to manipulate a large amount of spatial data. In this regard, the use of GIS has proven to be a powerful technique for analyzing and visualizing spatial data, which makes it an indispensable tool in the process of identifying potential sites for solar and wind energy projects.^{44,57} Nevertheless, Malczewski⁵⁸ stated that geo-information systems, on their own, are incapable of producing accurate results. On the other hand, MCDM models are used to rank the alternative decisions, given that all the involved criteria are taken into account.^{21,59} Hence, to further enhance decision-making processes related to site selection for alternative renewable energy solutions like solar and wind power installations or any other multiple factor-related problems, integrating GIS with MCDM is considered a valuable technique for solving problems of this type.^{11,20} This paper seeks to provide a preliminary assessment of qualified candidate sites that could host large-scale solar and wind installations in the region of Kasserine. To achieve this objective, a GIS-based SWARA-DEMATEL approach was utilized. Based on an in-depth review of the relevant academic literature and consultation with a knowledgeable group of experts, a comprehensive set of 12 decision criteria has been identified, which have been further categorized into three groups: Climatology, Topography, and Accessibility. The considered criteria

included global solar irradiance, temperature, wind speed, wind power density, percentage of windy days, slope, aspect orientation, land use, and proximity to residential areas, as well as grid and transport infrastructure. From a methodological perspective, the steps used in determining the optimal locations for solar and wind power plants are illustrated in Figure 1.

3.1 | SWARA approach

The SWARA methodology was initially developed by Keršulienė et al.⁶⁰ and has proven to be effective in weighting various multiattribute decision-making problems. In contrast to other methods like AHP and ANP, the SWARA approach offers a more direct means for experts' participation without having them engage in extensive pairwise comparisons or address concerns regarding consistency.^{61,62} This method involves evaluating and arranging the considered criteria based on their significance, from most important to least important, according to an expert's judgment. SWARA's key steps are explained as follows:

1. Rank the n criteria based on their significance.
2. Compute the importance scores ($\bar{A}_c$) obtained from T experts

$$\bar{A}_c = \sum_{t=1}^T \bar{A}_{c_t}. \quad (1)$$

3. Calculate the comparative importance S_j of the average value ($\bar{A}_c$) starting from the second ranked criterion, and determine its significance compared to the remaining criteria C_{j+1} .
4. Determine the coefficient K_j according to

$$K_j = \begin{cases} 1 & j = 1, \\ S_{j+1} & j > 1. \end{cases} \quad (2)$$

5. Determine the recalculated weight q_j

$$q_j = \begin{cases} 1 & j = 1, \\ \frac{q_{j-1}}{K_j} & j > 1. \end{cases} \quad (3)$$

6. Compute the final weight according to

$$W_j^S = \frac{q_j}{\sum_{k=1}^n q_k}. \quad (4)$$

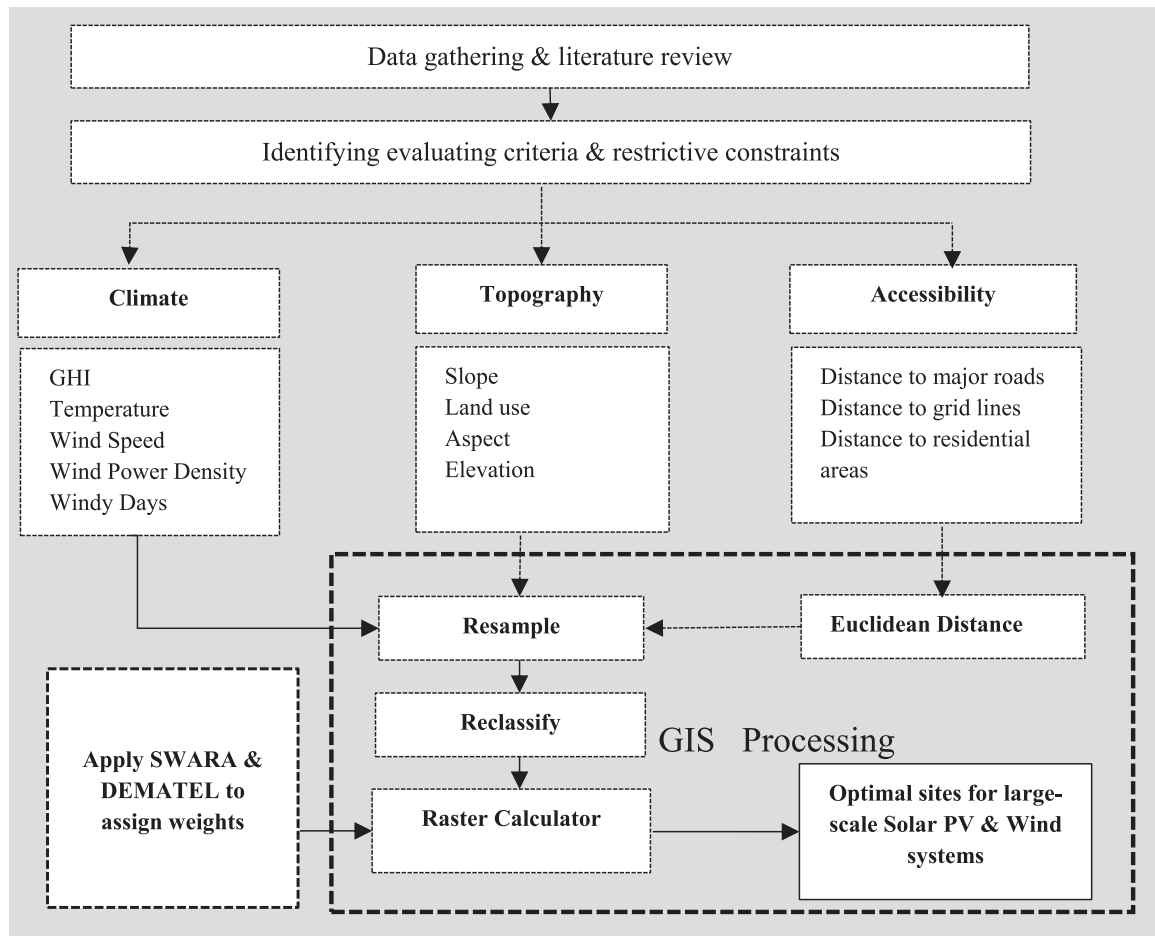


FIGURE 1 Conceptual steps for selecting optimal location for solar and wind systems. DEMATEL, Decision-Making Trial and Evaluation Laboratory; GHI, Global Horizontal Irradiance; GIS, Geographic Information System; SWARA, Step-wise Assessment Ratio Analysis.

3.2 | DEMATEL approach

The DEMATEL technique originated at the Battelle Geneva Institute in 1971 and is commonly employed to address intricate causal issues within complex systems. Its primary objective is to uncover relationships between various factors to determine direct and indirect dependencies among criteria.^{33,63,64} The foundation of the DEMATEL model lies in graph theory, which necessitates the use of an influential network relation map. By visually mapping out these relationships, it becomes simpler to grasp the significant factors and their causal implications within a complex problem structure. The following steps outline the key procedures involved in implementing the DEMATEL approach:

1. Create the direct comparison relation matrix for k experts and n criteria according to a 5-point scale from 0 (no influence) to 4 (very high influence)

$$Z = [Z_{ij}^k]. \quad (5)$$

2. Compute the aggregated matrix of k experts using the arithmetic mean
3. Normalize the aggregated matrix using the following equations:

$$L = \frac{1}{\max 1 \leq i \leq n \sum_{j=1}^n Z_{ij}}, \quad (6)$$

$$X = L \times Z. \quad (7)$$

4. Compute the total-relation matrix (T) as

$$T = X \times (I - X)^{-1}, \quad (8)$$

where I is the identity matrix

5. From the total matrix compute the sum of rows (R_i) and columns (C_i) according to the following equations:

$$R_i = \sum_{j=1}^n T_{ij} \quad \forall i, \quad (9)$$

$$C_i = \sum_{j=1}^n T_{ij} \quad \forall j. \quad (10)$$

6. Determine the DEMATEL weights as

$$W_j = \sqrt{(R_i + C_i)^2 + (R_i - C_i)^2}. \quad (11)$$

7. Compute the final weights as

$$W_j^D = \frac{W_j}{\sum_{k=1}^n W_k}. \quad (12)$$

The final weights for the overall ranking of the criteria associated with selecting the optimal locations for solar and wind systems are computed according to the following equation:

3.3 | Study area

Kasserine is one of the 24 Tunisian governorates, with a total area of 8260 km² and a population of 463,700 inhabitants. The region is a landlocked area lying between 35.25° north latitude and 8.7833° east longitude, in central-western Tunisia. It is bordered by El-Kef to the north, Algeria to the west, Gafsa to the south, and Sidi Bouzid and Siliana to the east (Figure 2). The region is characterized by a semi-continental climate, with cold and wet winters and relatively hot, dry summers. Agriculture remains the main economic activity in the region. The region has always been inflicted with long-standing challenges, namely unemployment, marginalization, and development disparities. These issues have resulted in economic and social inequalities, especially when compared to the more advantageous coastal regions.

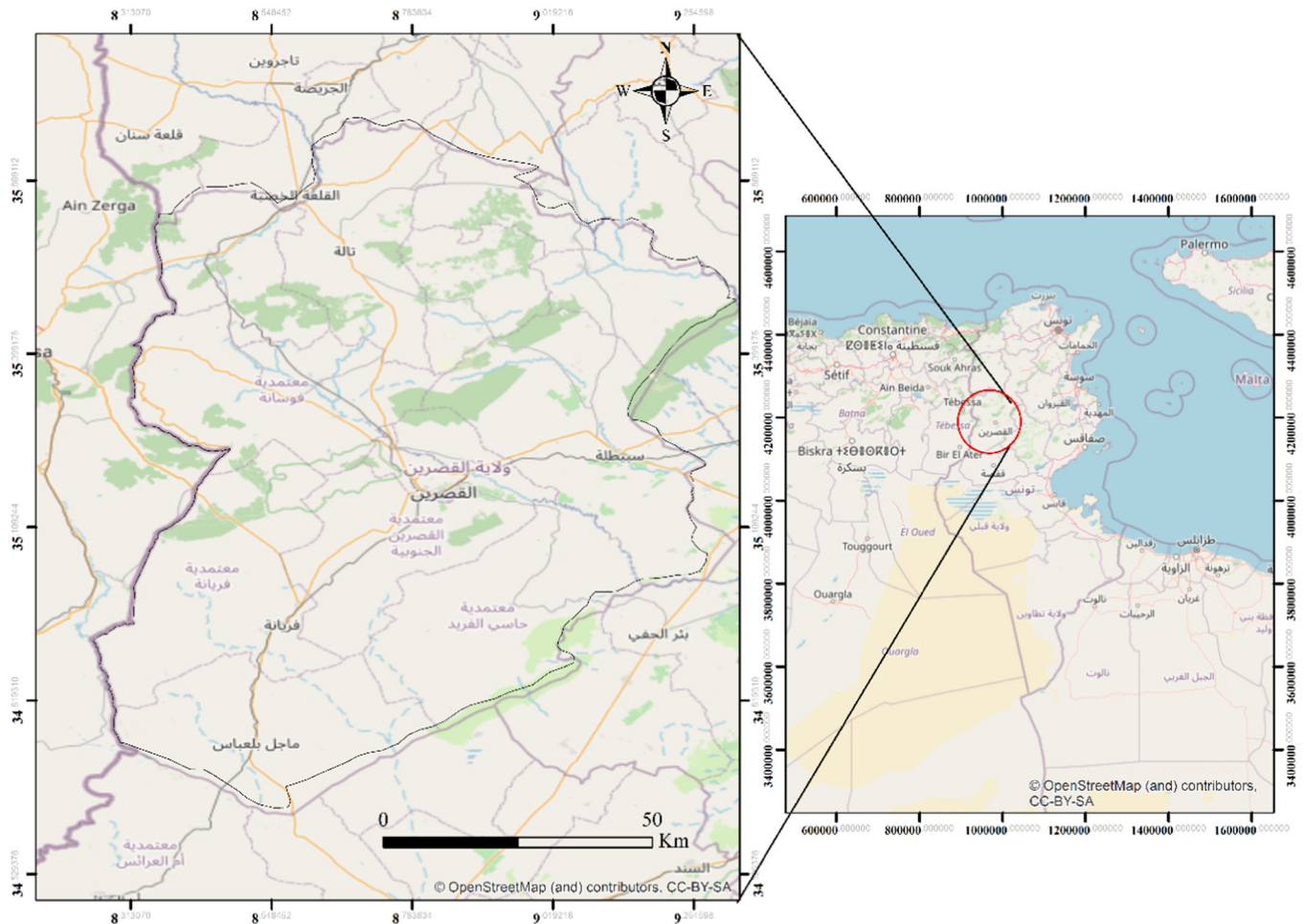


FIGURE 2 Study area.

3.4 | Identifying evaluating criteria

In recent years, there has been a significant amount of research focused on the use of MCDM techniques along with GIS to address the complexities associated with the siting of solar and wind facilities. In this regard, several studies have examined this topic in terms of land availability, resource assessment, environmental and social impacts, as well as infrastructure requirements.^{65,66} However, it is important to note that identifying potential sites within a specific area is heavily influenced by local factors and requires input from experts taking part in the decision-making process. Therefore, engaging highly qualified experts who possess in-depth knowledge about the energy status of a particular area is critical when conducting a multi-criteria decision analysis research activities.⁶⁷ In this study, focusing on the Kasserine region in central-western Tunisia as a real-life case study, a panel consisting of four well-informed experts was asked to provide feedback regarding the suggested criteria and express their opinions through the SWARA-DEMATEL approach. A brief profile of these experts can be found in Table A1.

3.4.1 | Climate criteria

A crucial factor in determining the ideal location for establishing a solar photovoltaic system is solar radiation, as it serves as the primary source of energy for PV panels. Hence, it is vital to ensure that the chosen site receives an adequate amount of sunlight throughout the year. Previous research indicates that a minimum threshold of 1300 kWh/year is necessary for the economic viability of a typical PV system.^{34,36}

However, it should be noted that the efficiency of PV modules depends on both isolation intensity and ambient temperature. Higher ambient temperatures negatively affect system performance. For every degree Celsius increase in cell temperature above 25°C, approximately 0.4%–0.5% of the generated energy is lost.³⁷ Hence, to optimize the performance of solar power plants, it is crucial to effectively manage wind speed and direction, as it helps lower the temperature of PV panels, which contributes significantly to their overall efficiency.³⁹

Similarly, when selecting an appropriate site for constructing a wind farm in a particular region, average wind speeds, wind power density, and the frequency of windy days are key factors to consider.^{33,34,41} Higher wind speeds falling within an optimal range indicate

TABLE 2 Climate criteria.

Main factor	Criteria	Attribute values	Rate
Climate	GHI (kWh/m ² /year)	<1299	1
		1300–1699	2
		1700–1899	3
		1900–1999	4
		>2000	5
	Wind Speed (m/s)	<4.99	1
		5–5.99	2
		6–6.99	3
		7–7.99	4
		>8	5
	Temperature (C°)	>23	1
		22–22.99	2
		21–21.99	3
		20–20.99	4
		<19.99	5
	Wind Power Density (W/m ²)	<100	1
		100–150	2
		150–200	3
		200–250	4
		>250	5
	Percentage of Windy Days	<30%	1
		30–40	2
		40–50	3
		50–60	4
		>60%	5

Abbreviation: GHI, Global Horizontal Irradiance.

favorable wind resource availability. Consequently, many studies have emphasized that wind speed stands out as a firsthand criterion. In this paper, wind speed, wind power density, and percentage of windy days were used for the wind analysis, while the criteria considered for conducting the solar analysis included solar irradiance (GHI) and temperature as well as wind speed, as shown in Table 2. A comprehensive data set incorporating detailed information on regional solar and wind patterns was obtained from the SOLARGIS portal (www.solargis.com), the Global Wind Atlas (www.globalwindatlas.info/area/Tunisia), and the National Institute of Meteorology (www.meteo.tn/fr/donnees-climatiques). These criteria are depicted in Figure 3A–E.

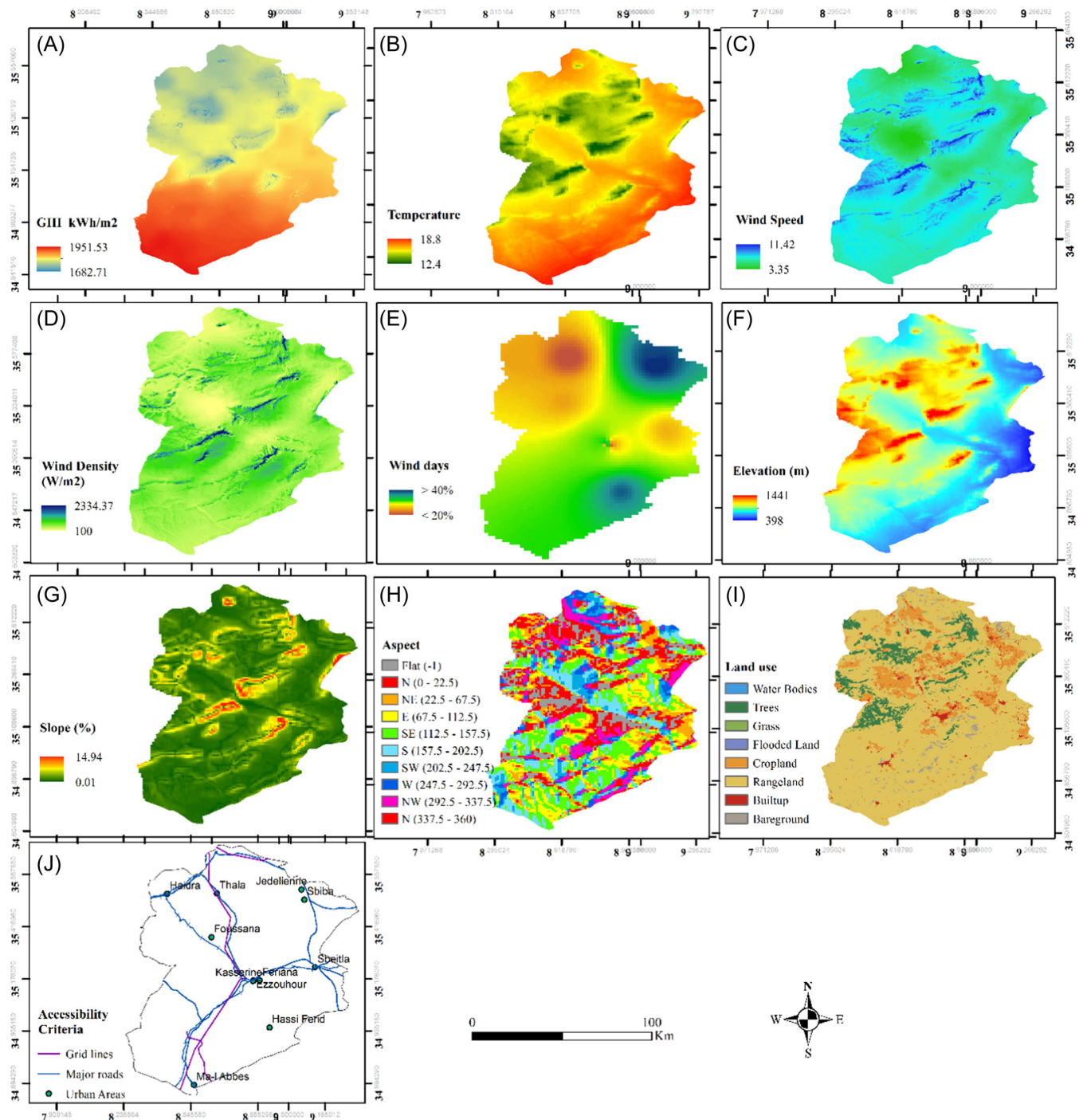


FIGURE 3 Criteria thematic maps. (A) Global Horizontal Irradiance (GHI), (B) ambient temperature, (C) wind speed, (D) wind power density, (E) percentage of windy days, (F) elevation, (G) slope, (H) aspect, (I) land use, (J) distance to grid lines, major roads, and urban areas.

3.4.2 | Topography criteria

The location of solar and wind facilities plays a critical role in their economic feasibility. To ensure cost-effectiveness, it is essential to choose sites that have flat terrain or gentle slopes. Steeper slopes can result in higher economic costs due to various factors, such as

increased construction and maintenance expenses. Additionally, these steep slopes may lead to the formation of shadows that negatively impact the performance of PV systems. Therefore, areas with gentler slopes are highly desirable for the establishment of such facilities.^{20,37} Typically, a slope threshold between 3% and 5% is considered acceptable for solar PV installations, while

wind power projects have varying ranges from 10% to 30%.³⁴ In addition, it is widely accepted that south-east to south-west-facing slopes in the northern hemisphere receive maximum sunlight exposure and are thus deemed most suitable.^{20,39}

Moreover, higher regions generally offer greater potential for both solar and wind energy resources compared to lower areas. However, increased altitude presents challenges in terms of transportation logistics and construction processes, which subsequently drive up project costs.^{25,39} The NASA 30 m Digital Elevation Model (DEM) was used to derive these criteria (Figure 3F–H). Furthermore, selecting an optimal site for a solar or wind power plant is subject to various limitations related to land use, which plays a crucial role in spatial planning. This constraint is prevalent in all site selection processes across different contexts. For instance, although favorable weather conditions may exist, unsuitable land use can hinder the feasibility of a solar or wind project.^{20,21} Thus, ideal locations should encompass areas free from significant constraints on land

utilization, such as mountains, sand dunes, forests, water bodies, military sites, etc. This criterion was obtained from a highly detailed map created by the European Space Agency with a resolution of 10 meters per pixel; this map is visualized in Figure 3I. The topography criteria used in this spatial analysis are detailed in Table 3. It is to note that the aspect criterion was not considered for the wind suitability analysis.

3.4.3 | Accessibility criteria

The proximity of solar and wind power plants to the grid is essential for their sustainability due to the direct relationship between distribution costs, power losses, and the distance from end-users. To enhance overall efficiency and significantly reduce power delivery expenses, it is advantageous to minimize line spacing between these facilities and the grid.²¹

Yet, transportation infrastructure is another key factor that has to be considered as it greatly impacts overall costs.^{68,69} The ease of access at a site helps decrease construction-related expenses during both the building phase and operational stages.⁷⁰ As such, it would not be recommended to deploy these facilities in areas with challenging accessibility. These criteria are shown in Table 4, and their thematic maps are illustrated in Figure 3J.

TABLE 3 Topography criteria.

Main factor	Criteria	Attribute values	Rate
Topography	Slope (°)	>10	1
		8–10	2
		5–8	3
		2–5	4
		<2	5
	Land use	Built-up, Water bodies, Forests, etc.	1
		Cropland	2
		Shrubland	3
		Sparse Vegetation	4
		Bare Lands	5
	Aspect	Flat	1
		N, NE, NW	2
		E, W	3
		SE, SW	4
		S	5
Elevation (m)		<100	1
		100–150	2
		150–200	3
		200–250	4
		>250	5

TABLE 4 Accessibility criteria.

Main factor	Criteria	Attribute Values	Rate
Accessibility	Dist. to Roads (km)	>15	1
		10–15	2
		5–10	3
		1–5	4
		0.5–1	5
	Dist. To Power lines (km)	>15	1
		10–15	2
		5–10	3
		1–5	4
		0.3–1	5
	Dist. To Residential Areas (km)	20–25	1
		15–20	2
		10–15	3
		5–10	4
		2–5	5

TABLE 5 Restrictive constraints.

Restrictive layer	Boolean algebra restriction
Distance from protected areas	$x < 0.5\text{ km}$
Distance from grid	$x < 0.3\text{ km}$ and $x > 20\text{ km}$
Distance from roads	$x < 0.5\text{ km}$ and $x > 20\text{ km}$
Distance from residential areas	$x < 2\text{ km}$ and $x > 25\text{ km}$
Slope	$x > 10\%$
Landuse	$x \neq$ Bare ground, shrubland, and medium grassy vegetation

3.5 | Restrictive constraints

To assess the suitability of different areas for the development of renewable energy power plants, a series of constraints have to be taken into consideration.^{20,21} In this research work, the considered constraints (Table 5) were determined through an exhaustive literature survey. The Boolean algebra within the ArcGIS 10.8 software was used to generate and aggregate those factors into one single layer. From this aggregated layer, cells with a value of “1” indicate the absence of limitations, and therefore it is possible to construct renewable facilities. In contrast, cells with a value of “0” refer to the presence of restrictions, which means the installation of such facilities is not possible. These restrictive constraints are illustrated in Figure 4.

4 | RESULTS AND DISCUSSION

4.1 | SWARA-DEMATEL results

Using the SWARA method, experts were tasked with ranking criteria based on their level of significance (see Tables B1 and B2). The similarity in assigned ranks is notable evidence of a strong consensus among experts regarding their preferences for the established criteria. The overall rankings are determined through an arithmetic average calculation of individual expert ranks. Next, the comparative importance averages S_j are determined, starting from the second-ranked criterion onwards. This shows how significant one particular criterion (c_j) compared with its successor (c_{j+1}). Then, coefficient values k_j and recalculated weight values q_j for each criterion are computed using Equations (2) and (3). Lastly, the ultimate weights for each criterion can be derived from these q_j values according to Equation (4) (see Table 6).

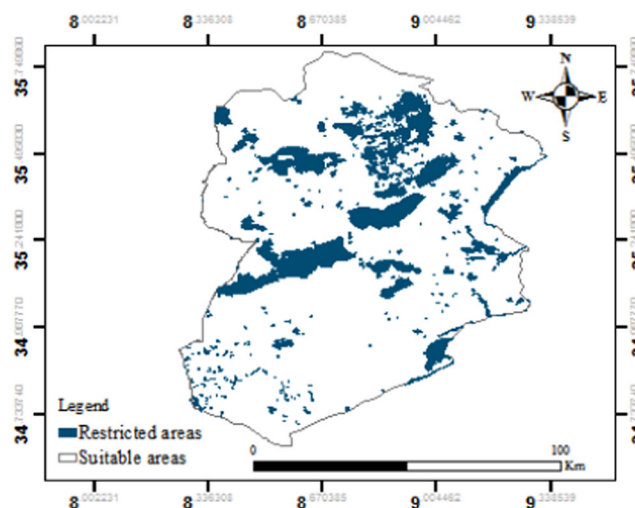


FIGURE 4 Restrictive constraints map.

In the second stage, the experts were again invited to assess the same criteria using a 5-point scale. The aggregated initial matrices were created based on Equation (5) (see Tables C1 and C2). Then, the total relationship matrix was formulated through Equations (6)–(8). Then, the DEMATEL rankings were computed on the basis of $(R_i + C_i)$ and $(R_i - C_i)$ as illustrated in Table 7. From the DEMATEL approach, it was observed that the main factors of climate and topography were in the cause group ($R_i + C_i > 0$), while accessibility was in the effect group. This indicates the prime importance of the availability of resources along with the topographic features of the location in the decision-making process (Figure 5). On the other hand, in terms of decision criteria, it was found that wind speed, distance to residential areas, distance to major transport links, GHI, wind speed, slope, and land use were the main causative criteria for both solar and wind (Figure 6). This implies that adequate transportation infrastructure, solar and wind resources, a proper type of land, and proximity to urban areas have to be thoroughly investigated to ensure the feasibility of the selected site.

To assess the extent of significance for the identified criteria, weights were computed through Equation (13) and then normalized, as demonstrated in Table 8. The results obtained indicate that experts place greater emphasis on factors associated with resource availability and topography, such as solar radiation, wind speed, and slope. Conversely, less attention is given to elevation and proximity to residential areas. Moreover, for solar technology, it was observed that the level of importance placed on criteria varies significantly; specifically, solar irradiance, slope, and wind speed had the highest influential weights at 29%, 13.08%, and 12.95%, respectively. Yet transportation infrastructure and temperature

TABLE 6 SWARA results.

Solar PV				Onshore wind			
Criteria	K_j	q_j	Weights	Criteria	K_j	q_j	Weights
C ₁	-	1.000	0.536	C ₁	-	1.000	0.536
C ₁₁	-	1.000	0.529	C ₁₁	-	1.000	0.470
C ₁₂	1.788	0.559	0.296	C ₁₂	1.850	0.541	0.254
C ₁₃	1.688	0.332	0.175	C ₁₃	1.700	0.588	0.276
C ₂	1.68	0.323	0.291	C ₂	1.68	0.323	0.291
C ₂₁	-	1.000	0.516	C ₂₁	-	1.000	0.516
C ₂₂	1.725	0.580	0.299	C ₂₂	1.725	0.580	0.299
C ₂₃	1.625	0.357	0.185	C ₂₄	1.625	0.357	0.185
C ₃	1.84	0.543	0.173	C ₃	1.84	0.543	0.173
C ₃₁	-	1.000	0.422	C ₃₁	-	1.000	0.492
C ₃₂	1.656	0.604	0.253	C ₃₂	1.656	0.604	0.297
C ₃₃	1.413	0.427	0.180	C ₃₃	1.413	0.427	0.211
C ₃₄	1.238	0.345	0.145	-	-	-	-

TABLE 7 DEMATEL results.

Solar PV						Onshore wind					
Solar criteria	R_i	C_i	$R_i + C_i$	$R_i - C_i$	Weight	Wind Criteria	R_i	C_i	$R_i + C_i$	$R_i - C_i$	Weight
C ₁	2.967	3.343	6.309	-0.38	0.348	C ₁	2.967	3.343	6.309	-0.38	0.348
C ₁₁	5.97	5.43	11.40	0.54	0.365	C ₁₁	1.862	0.718	2.580	1.143	0.368
C ₁₂	4.15	4.93	9.08	-0.78	0.292	C ₁₂	1.215	0.945	2.160	0.270	0.284
C ₁₃	5.49	5.25	10.73	0.24	0.343	C ₁₃	0.429	1.843	2.272	-1.41	0.349
C ₂	2.324	3.254	5.578	-0.93	0.311	C ₂	2.324	3.254	5.578	-0.93	0.311
C ₂₁	14.320	13.842	28.162	0.478	0.343	C ₂₁	14.32	13.842	28.162	0.478	0.343
C ₂₂	12.851	14.389	27.240	-1.54	0.332	C ₂₂	12.851	14.389	27.240	-1.54	0.332
C ₂₃	13.893	12.834	26.728	1.059	0.325	C ₂₃	13.893	12.834	26.728	1.059	0.325
C ₃	3.677	2.371	6.048	1.307	0.341	C ₃	3.677	2.371	6.048	1.307	0.341
C ₃₁	1.722	0.525	2.247	1.722	0.525	C ₃₁	1.959	1.010	2.969	0.949	0.370
C ₃₂	1.472	0.595	2.067	1.472	0.595	C ₃₂	1.703	1.053	2.756	0.650	0.336
C ₃₃	0.259	1.445	1.704	0.259	1.445	C ₃₃	0.141	1.739	1.880	-1.60	0.293
C ₃₄	0.392	1.279	1.671	0.392	1.279	-	-	-	-	-	-

were other important parameters, with scores of 9.47% and 9.07%, respectively. Similarly, wind speed, slope, percentage of windy days, and wind power density were perceived as the most significant factors for installing wind power plants, with relative weights of 25.82%, 17.6%, 14.41%, and 10.77% in respective order. On the other hand, elevation and proximity to residential areas were regarded as the least influential factors for both technologies.

4.2 | Spatial analysis

In this comprehensive study, several steps were undertaken to ensure a thorough analysis of the spatial data. Initially, rescaling, resampling, and reclassifying techniques were employed on the diverse input layers to standardize their values and make them compatible for further analysis. The final suitability maps were generated based on the adopted model and using the raster

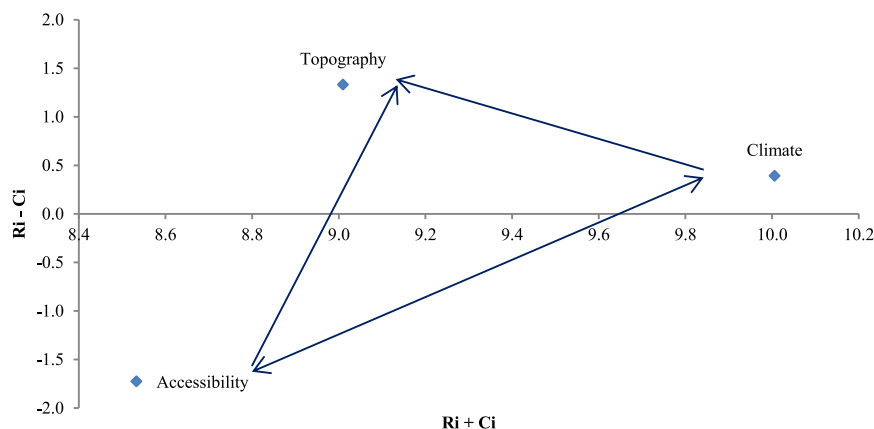


FIGURE 5 The cause-and-effect diagram of the main factors.

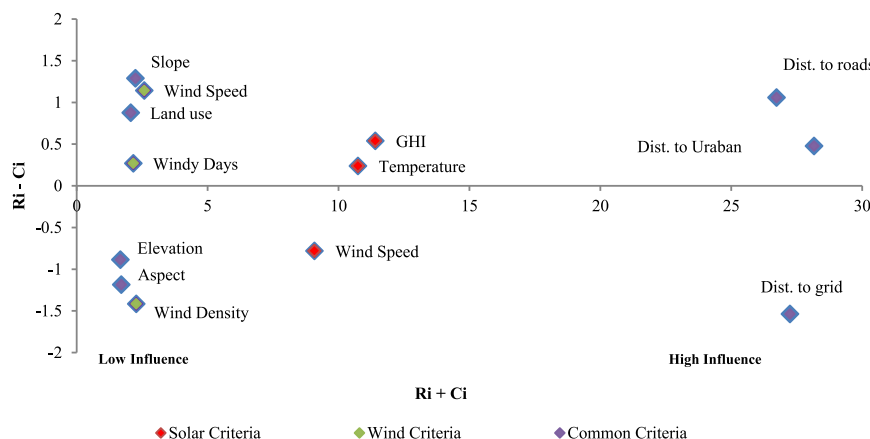


FIGURE 6 The cause-and-effect diagram of the decision criteria.

TABLE 8 Final normalized weights for solar and wind.

Factor	Criteria	Relative weight	
		Solar PV	Onshore wind
Climate	GHI	29.0% (1)	-
	Wind speed	12.95% (3)	25.82%
	Wind Power density	-	10.77%
	Windy Days	-	14.41%
	Temperature	9.03% (5)	-
Accessibility	Distance to transport links	9.47% (4)	9.47%
	Distance to grid network	5.31%	5.31%
	Distance to urban areas	3.20%	3.20%
Topography	Slope	13.08% (2)	17.60%
	Land use	8.96% (6)	8.70%
	Aspect	5.18%	-
	Elevation	3.81%	4.70%

Abbreviation: GHI, Global Horizontal Irradiance.

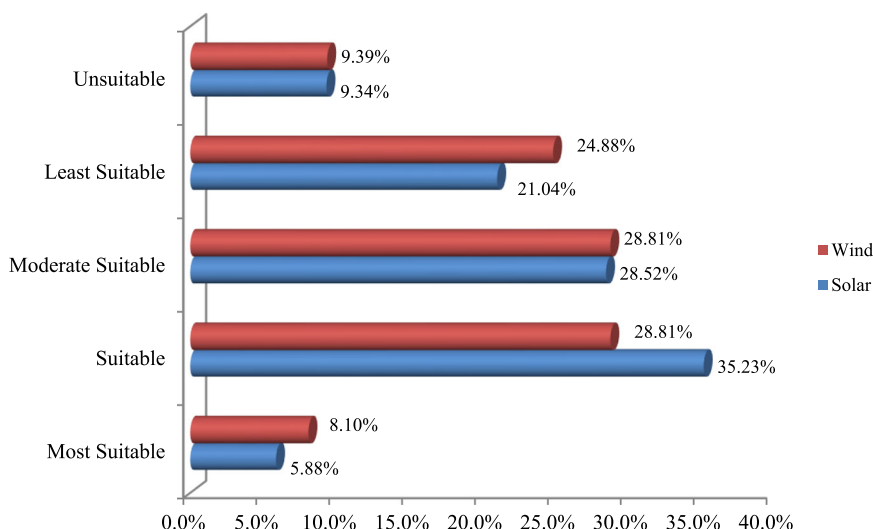
calculator tool. These maps consisted of five classes, representing the suitability levels for solar and wind energy development. Class 5 denoted the “most suitable” sites, while Class 1 indicated areas that were deemed “unsuitable.”

To further refine and identify optimal locations for large-scale solar PV and wind facilities within these suitable areas, a minimum threshold criterion of at least 1 km² was applied. This ensured that only regions with sufficient land availability would be considered as potential candidates for establishing these systems.

4.3 | Potential sites

The analysis of the potentially viable areas for solar energy production revealed that the “most suitable” areas covered 467.6 km², representing 5.88% of the total available land area. Meanwhile, a significant portion of about 35.23% fell under the category of “suitable,” 28.52% for “moderately suitable,” 21.04% for “least suitable,” and 9.34% for “unsuitable,” respectively (Figure 7). When examining the distribution of these highly suitable sites across the Kasserine region, it became apparent that the

FIGURE 7 Final land suitability distribution.



most concentrated clusters were observed in the southern parts, most notably in Majel Belabbes, Feriana, and Kasserine Sud (Figure 8A).

Conversely, wind suitability exhibited different patterns compared to solar locations. The map displayed an allocation where only around 8.10% (635 km²) constituted “highly suitable” sites out of the total surface area analyzed. Locations falling within categories labeled as “suitable,” “moderately suitable,” “least suitable,” and “unsuitable” accounted respectively for 28.81%, 28.81%, 24.88%, and 9.39% (Figure 7). Moreover, sites of high suitability were densely scattered across the southern, central, and most northern parts of the Kasserine region. Among those sites, Majel Belabbes, Kasserine Sud, Feriana, and Thala appeared to be by far the most favorable candidate locations for installing wind facilities (Figure 8B). Statistically speaking, considering these highly suitable areas for both technologies, it would be possible to generate an annual output power of 7788 GWh for wind and 94386 GWh for solar, respectively, which is equivalent to more than a third and fourfold the entire demand as of 2021.¹³

Furthermore, the integrated approach enabled the identification of potential locations that are capable of hosting both technologies. By analyzing the geographical overlap of suitable locations, it was possible to determine 349 km² that are well-suited for solar-wind hybrid systems. In terms of extensive availability land-wise, with a wind speed of 8.06 m/s, a wind power density of 320 W/m², and an annual GHI of 1942 kWh/m², Majel Belabbes makes up the ideal location for such systems, accounting for nearly 65% of those total sites (Figure 8C).

The spatial analysis conducted in this study can provide valuable guidance to policymakers in selecting the most appropriate locations for large-scale solar PV, onshore wind, and solar-wind hybrid projects in the

Kasserine region. The Tunisian government has recently announced plans to construct three PV projects with a capacity of 10 MW each under the authorization regime. These projects (two in Feriana and one in Kasserine Sud) are poised to take place at the end of 2023. Interestingly, both locations were in line with the results obtained, as the “most suitable” sites in these areas represented 26.28% (Feriana) and 16.07% (Kasserine Sud), respectively.

In addressing key research questions on the integration of renewable energy technologies in the Kasserine region, several key findings emerged. First, the development of renewable energy projects such as large-scale solar PV and wind systems is likely to lead to direct and indirect job creation.⁷¹ Direct jobs could arise from the construction, operation, and maintenance of renewable energy installations. Indirect jobs could be generated in supporting industries such as the manufacturing of components, transportation, and services required for the ongoing operation of energy facilities. For a region characterized by economic challenges, such as Kasserine, this could contribute substantially to local employment and skill development.

In general terms, renewable energy projects like solar PV and wind energy can offer several financial benefits over conventional energy sources. Once installed, solar and wind energy systems typically have very low operating and maintenance costs compared to conventional power plants. In addition, the costs associated with producing energy from solar and wind sources are highly predictable, especially as there are no fuel costs subject to market fluctuations. This can lead to more stable energy prices in the long term.^{72–74}

Second, Tunisia’s geopolitical and social situations influence investment security and partnership opportunities. In addition, the legislative landscape shapes the

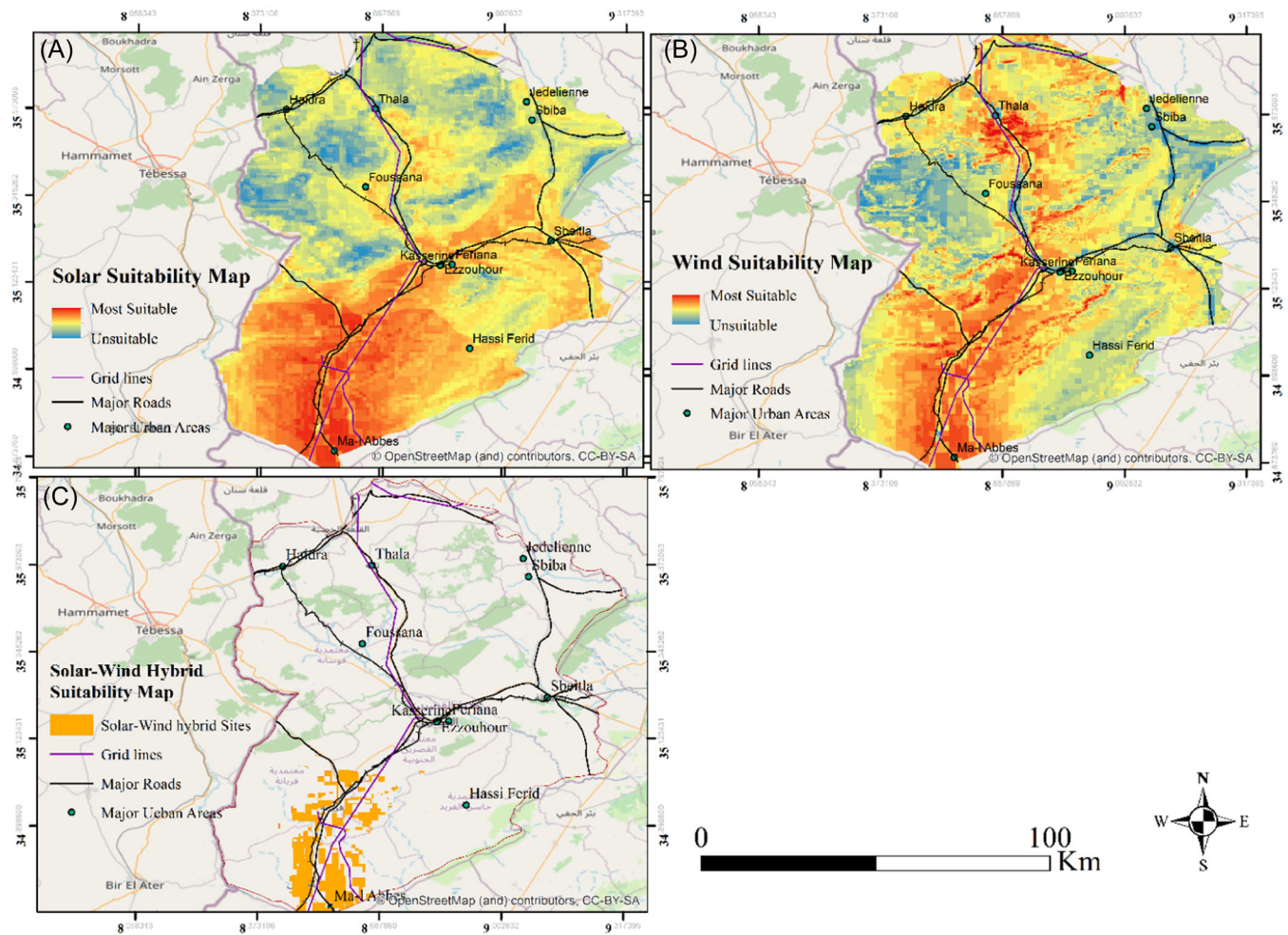


FIGURE 8 Final Suitability maps. (A) Solar suitability map, (B) wind suitability map, and (C) solar-wind hybrid systems suitability map.

practicalities of renewable energy projects. Existing laws and regulations related to land use, the environment, and energy production set the ground rules for where and how renewable energy projects can be deployed.⁷⁵ Moreover, legislation regarding the connection to and use of the national grid is crucial for renewable energy projects. This impacts how easily generated energy can be distributed and sold.⁷⁶ Besides, community engagement and participation in the planning and decision-making process tend to increase acceptance of such facilities, as effective communication can help build trust and resolve any possible conflicts.^{77,78} Therefore, socio-political aspects must align favorably to enable the successful deployment of renewable energy systems at the identified sites.

Lastly, the renewable energy strategy devised for this region can be used as a model for other similar regions in Tunisia and neighboring countries. This approach, which harnesses sophisticated tools like GIS for mapping and SWARA-DEMATEL for multicriteria

decision-making, has successfully identified optimal locations for solar photovoltaic and wind energy projects in Kasserine. Since these methods take into account local climatic, topographical, and infrastructural factors, they can be tailored to other regions. By following this example, policymakers and developers in other areas with comparable geographical and socioeconomic profiles can implement similar strategies to assess and unlock their renewable energy potential, contributing to regional development and sustainability goals.

5 | DISCUSSIONS

Considering the commonalities in solar and wind site selection decision-making problems, such as the abundance of resources, topographic features, and necessary infrastructure, our findings align well with those in the literature. This research paper and others in the

TABLE 9 Comparison with similar studies in neighboring countries.

	Reference	Most suitable area (km ²)			Methodology
		Solar PV	Wind	Hybrid	
This study		467.6	635	349	GIS-SWARA DEMATEL
Igdir, Turkey	[39]	524.5	147.2	49.1	GIS-AHP
Tehran, Iran	[81]	1477.28	2680.7	-	GIS-AHP
Misurata, Libya	[21]	1667	-	-	GIS-SWA'TEL
Marrakesh-Safi, Morocco	[38]	9518	-	-	GIS-AHP
Kayseri, Turkey	[82]	-	-	2080	GIS-AHP
Dakhla region -Morocco	[83]	-	-	198	GIS-FAHP
Assiut, Egypt	[34]	20 sites (50.19–98.05)	723	66.875	GIS-AHP

literature involve experts in the decision-making process, as their knowledge contributes to the weighting and assessment of the various criteria. All studies aimed at providing a thorough evaluation of the physical and geographical possibilities for these renewable installations. The findings across the literature offer critical insights that are beneficial for policymakers and planners, providing a decision-support tool that can guide the selection of renewable energy sites with an eye toward sustainability and strategic development. The prime focus on this specific region is driven by the fact that Kasserine is often labeled as the most marginalized part of the country. Thus, deploying solar and wind facilities would spur economic growth across different sectors like construction, operations, maintenance, and local support services. Interestingly, Kasserine has been recognized as having substantial solar and wind potential, which presents promising opportunities for policymakers and investors.^{79,80} These findings are particularly significant when compared to similar studies conducted in other regions in terms of the available area and estimated generated energy for either single or hybrid systems, as illustrated in Table 9. This particular study focuses on exploring solar PV and wind as well as solar-wind hybrid systems in the Kasserine region, taking into account the unique social, political, and investment challenges in Tunisia. Other studies concentrated solely on solar, wind or examined additional renewable technologies such as bioenergy or geothermal by emphasizing different barriers, like technical maturity, market dynamics, and policy frameworks, reflecting the unique challenges faced in each region.

However, despite the ambitious goal set by the Tunisian government to generate 30% of its electric power from renewable sources by 2030, a move aimed at achieving energy independence, diversifying the energy mix, reducing fossil fuel imports, and lowering emission levels, there have been limited signs of progress. Various factors, such as social and political challenges, the lack of adequate funding, and an unfavorable investment climate, have impeded the pace of implementation of this plan.

In light of the fact that Kasserine is one of the most economically and socially marginalized regions of the country, it is imperative that concrete measures be taken to encourage investment in solar and wind energy projects to uplift and promote the development of this least-privileged region.^{84,85} The process of identifying optimal locations typically entails not only identifying suitable areas but also improving grid infrastructure, transportation systems, manufacturing facilities, and educational and training centers.⁸⁶ By investing in these essential aspects alongside renewable energy projects, significant improvements can be made to social welfare and living standards within this designated region.^{7,85,87}

6 | CONCLUSION

The increasing prominence of sustainable and eco-friendly electricity generation has led to the emergence of solar and wind power as viable options. As this sector continues its rapid expansion, it is imperative to identify appropriate locations for energy production to overcome

challenges stemming from inadequate policy support and regulatory frameworks. Consequently, the primary objective of this study was to carry out a comprehensive spatial analysis regarding the suitability of land in Tunisia for establishing large-scale solar PV and wind power plants in the region of Kasserine, central-western Tunisia. To do so, we focused on applying a GIS-based SWARA-DEMATEL integrated approach to determine the most feasible locations for harnessing solar and wind energy in this region. The adopted model took into account various factors such as solar radiation intensity, wind speed and wind power density, land use, grid and transportation infrastructure accessibility, and environmental constraints. By integrating and analyzing these spatially explicit data layers, the approach provided a comprehensive evaluation of potential sites for solar and wind energy as well as solar-wind hybrid projects. Based on the results obtained, the region of Kasserine has emerged as a prime location for both technologies, thanks to its favorable climatic and geographical characteristics. To capitalize on this enormous solar and wind potential within this region, well-designed policy tools are crucial in accelerating the shift to renewable energy systems. Measures such as feed-in tariffs, tax exemptions, and renewable portfolio standards provide financial incentives for investment in clean energy projects. Additionally, streamlined permitting processes and grid integration policies enable smoother adoption of these systems. Collaborative partnerships between government, industry, and research institutions further facilitate knowledge sharing, innovation, and capacity-building. These policy tools create a favorable environment for renewable energy growth and help drive the transition toward a more sustainable future. By addressing the key research questions and considering the exploitable areas for either single or hybrid systems, it is evident that the shift toward green technologies in Kasserine is very likely, provided that adequate policies and concrete measures are taken.

By developing resilient social, ecological, and technological systems through the adoption of green systems, this region can achieve a multitude of SDGs simultaneously. Embracing these technologies would not only ensure access to affordable and clean energy (SDG 7) but also create employment opportunities in the renewable energy sector, promote economic growth (SDG 8), mitigate carbon footprints, and combat climate change as well (SDG 13). Moreover, investing in green technologies improves energy efficiency and infrastructure, leading to more sustainable cities and communities (SDG 11). It also fosters responsible consumption and production patterns by reducing reliance on fossil fuels (SDG 12).

Nonetheless, limitations of this present study include the exclusion of other criteria such as average cloudy days, soil texture, lightning strike flash rate, land ownership, and policy regulations, which can have a substantial impact on the economic viability and implementation of these systems. Moreover, installing solar and wind facilities requires significant upfront costs compared to conventional power plants. Additionally, owing to their intermittent nature, storage solutions or backup systems are required to ensure a constant supply. Furthermore, integrating large amounts of renewable energy can require upgrades to the existing grid infrastructure, which can potentially increase the overall costs. As such, a detailed financial assessment is needed, including capital expenditures (CAPEX), operational expenditures (OPEX), internal rates of return (IRR), and levelized cost of electricity (LCOE). Such an analysis would take into account local conditions, energy policies, and market dynamics to compare the investment returns of renewable energy sites to those of other energy sources in Tunisia.

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APPENDIX A

Table A1.

TABLE A1 Profile of experts.

	Designation	Organization	Area of experience	Experience
1	Academician	l'Ecole Nationale d'Ingénieurs de Tunis (ENIT)	Renewable energies	38
2	General Director	ANME	Solar and wind energy policies	27
3	Senior Engineer	STEG	Solar and wind energy	27
4	General Director	Alcor research and consulting firm	PhD, economist specializing in energy and climate change	25

APPENDIX B

Tables B1 and B2.

Criteria	Denotation	Expert 1	Expert 2	Expert 3	Expert 4	Average
Climate	C1	-	-	-	-	-
GHI	C11	-	-	-	-	-
Temperature	C12	0.90	0.75	0.70	0.80	0.788
Wind speed	C13	0.80	0.65	0.60	0.70	0.688
Accessibility	C2	0.75	0.80	0.65	0.50	0.675
Dist. to roads	C21	-	-	-	-	-
Dist. to grid lines	C22	0.80	0.75	0.65	0.70	0.725
Dist. to urban areas	C23	0.70	0.65	0.55	0.60	0.625
Topography	C3	0.90	0.85	0.75	0.85	0.838
Slope	C31	-	-	-	-	-
Land use	C32	0.70	0.60	0.78	0.55	0.656
Aspect	C33	0.45	0.40	0.50	0.30	0.413
Elevation	C34	0.30	0.20	0.25	0.20	0.238

Abbreviation: GHI, Global Horizontal Irradiance.

TABLE B1 Average comparative importance for solar PV criteria.

TABLE B2 Average comparative importance for solar PV criteria.

Criteria	Denotation	Expert 1	Expert 2	Expert 3	Expert 4	Average
Climate	C1	-	-	-	-	-
GHI	C11	-	-	-	-	-
Temperature	C12	0.70	0.75	0.65	0.70	0.700
Wind speed	C13	0.90	0.85	0.90	0.85	0.875
Accessibility	C2	0.75	0.80	0.65	0.50	0.675
Dist. to roads	C21	-	-	-	-	-
Dist. to grid lines	C22	0.80	0.75	0.65	0.70	0.725
Dist. to urban areas	C23	0.70	0.65	0.55	0.60	0.625
Topography	C3	0.90	0.85	0.75	0.85	0.838
Slope	C31	-	-	-	-	-
Land use	C32	0.70	0.60	0.78	0.55	0.656
Elevation	C33	0.45	0.40	0.50	0.30	0.413

Abbreviation: GHI, Global Horizontal Irradiance.

APPENDIX C

Tables C1 and C2.

TABLE C1 Aggregated initial matrix for solar PV.

	C ₁	C ₁₁	C ₁₂	C ₁₃	C ₂	C ₂₁	C ₂₂	C ₂₃	C ₃	C ₃₁	C ₃₂	C ₃₃	C ₃₄
C ₁	0.00				2.33				1.33				
C ₁₁		0.00	2.33	1.33									
C ₁₂		2.67	0.00	2.33									
C ₁₃		3.00	3.67	0.00									
C ₂	3.67				0.00				3.00				
C ₂₁						0.00	3.67	3.00					
C ₂₂						2.00	0.00	1.33					
C ₂₃						2.67	2.33	0.00					
C ₃	2.67				3.67				0.00				
C ₃₁										0.00	3.67	3.33	3.67
C ₃₂										3.00	0.00	2.67	3.00
C ₃₃										0.33	0.00	0.00	1.33
C ₃₄										0.00	0.33	2.67	0.00

TABLE C2 Aggregated initial matrix
for wind.

	C ₁	C ₁₁	C ₁₂	C ₁₃	C ₂	C ₂₁	C ₂₂	C ₂₃	C ₃	C ₃₁	C ₃₂	C ₃₃
C ₁	0.00				2.33				1.33			
C ₁₁		0.00	3.67	3.00								
C ₁₂		1.00	0.00	3.67								
C ₁₃		1.00	0.00	0.00								
C ₂	3.67				0.00				3.00			
C ₂₁						0.00	3.67	3.00				
C ₂₂						2.00	0.00	1.33				
C ₂₃						2.67	2.33	0.00				
C ₃	2.67				3.67				0.00			
C ₃₁										0.00	3.67	3.33
C ₃₂										3.00	0.00	2.67
C ₃₃										0.33	0.00	0.00